DRIVER DROWSINESS DETECTION SYSTEM

Seminar Report Submitted by

PRINTO MATHEW

Register No: 220021087299

In partial fulfilment of the requirements for the award of the degree

Of

BACHELOR OF COMPUTER APPLICATION (BCA)

P.G.M COLLEGE

KANGAZHA

MAHATMA GANDHI UNIVERSITY

KOTTAYAM

2022 – 2025

DEPARTMENT OF COMPUTER APPLICATIONS

P.G.M COLLEGE

KANGAZHA

CERTIFICATE

Certified that this Seminar report **“ *D*river Drowsiness Detection System”** *is the bona fide work of* **Mr. PRINTO MATHEW** *with Register Number:* **220021087299** *who carried out the work under my supervision. Certified further, that to the best of my knowledge the work reported herein does not form part of any other seminar report or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.*

Head of the Department	**Seminar Co-ordinator**	**Internal Guide**
Mrs. Bhagya M.U	Mrs. Bhagya M.U	Mrs. Chithra B

Submitted for the viva-voce held on...

External Examiner

<u>DECLARATION</u>

I hereby declare that the seminar report "**Driver Drowsiness Detection System**" is a bona fide work done at PGM College Devagiri, Kangazha towards the partial fulfilment of the requirements for the award of the Degree of Bachelor of Computer Applications (BCA) from Mahatma Gandhi University, Kottayam, during the academic year 2022-2025.

Date :…………….	**PRINTO MATHEW**
Place: Kangazha	**Reg No:220021087299**

ACKNOWLEDGEMENT

First and foremost I thank God Almighty for His blessings for this seminar. I take this opportunity to express my gratitude to all those who have helped me in completing the seminar successfully.
It has been said that gratitude is the memory of the heart. I wish to express my sincere gratitude to our Principal **Prof. JEEN ABHRAHAM** for providing good faculty for guidance.

I owe a great debt of gratitude towards our head of the department **Mrs. BHAGYA M.U** for helping us. I am thankful to our seminar guide **Mrs. CHITHRA B** for her proper guidance throughout the seminar. I also express my gratitude towards the supporting staffs for technical assistance.

I am indebted to my beloved teachers whose cooperation and suggestions throughout the seminar which helped me a lot. I also thank all my friends and classmates for their interest, dedication and encouragement shown towards the seminar.

I convey hearty thanks to parents for the moral support, suggestion and encouragement to make this venture a success.

PRINTO MATHEW

ABSTRACT

The Drivers who do not take regular breaks when driving long distances run a high risk of becoming drowsy, a state which they often fail to recognize early enough according to the experts. According to research it is proven that more than one quarter of the road accidents happen due to the drowsiness of the driver which means that more accidents are caused due to tired drivers rather than the drivers who are drunk. In state of extended speed range, a system can notify drivers about their inattentiveness and state of fatigue with the help of an alarm. There are many such alarm systems which are focusing on alerting the driver because accidents caused due to human error are increasing day by day and causing many deaths and injuries worldwide. The main reasons that cause fatal crashes and highway accidents these days are the drowsiness and sleeping of the driver. Driver drowsiness detection system is a safety technology that can prevent accidents that are caused by drivers who fell asleep while driving. This system will alert the driver when the drowsiness is detected and hence prevent the accidents from taking place.

The rising prevalence of road accidents caused by driver fatigue highlights the urgent need for effective detection systems. The Driver Drowsiness Detection System leverages advancements in image processing, machine learning, and facial analysis to identify early signs of drowsiness, such as prolonged eye closure and yawning. By employing technologies like OpenCV and dlib, the system analyzes real-time video to monitor facial landmarks and detect critical behavioral patterns. This seminar will explore the integration of behavioral, vehicle-based, and physiological measures in drowsiness detection systems, with a focus on overcoming challenges such as environmental lighting and accuracy. The proposed system aims to enhance road safety by providing timely alerts to prevent fatigue-related accidents.

CONTENTS

1. INTRODUCTION

Driver drowsiness is a leading cause of road accidents worldwide, often resulting in catastrophic consequences such as injuries, fatalities, and property damage. According to studies, a significant proportion of accidents are attributed to drowsy drivers, surpassing the rate of incidents caused by drunk driving. Long-distance and commercial drivers, such as those operating buses, taxis, and trucks, are particularly vulnerable due to extended driving hours and inadequate rest.

The Driver Drowsiness Detection System aims to address this critical issue by utilizing cutting-edge technologies in image processing and computer vision. This system monitors the driver in real-time, detecting signs of fatigue such as prolonged eye closure, reduced blinking rate, and yawning. By analyzing facial landmarks using tools like OpenCV and dlib, the system can accurately identify drowsy behavior and provide timely alerts to the driver.

The primary goal of this technology is to enhance road safety by preventing accidents before they occur. Unlike traditional methods, this system focuses on non-invasive and cost-effective techniques, making it practical for widespread adoption. Additionally, the system overcomes challenges such as poor lighting conditions by integrating advanced features like infrared-based detection.

As human errors remain a significant factor in road accidents, the development and implementation of such innovative solutions are crucial. This seminar not only addresses a pressing issue but also contributes to the broader mission of reducing fatalities and improving the overall safety of roadways.

2. WHAT IS AI?

Artificial intelligence (AI) is a collection of technologies that allow computers to perform tasks that normally require human intelligence. It refers to the simulation of human intelligence in machines that are programmed to think, learn, and perform tasks typically requiring human cognition. These machines can analyze data, recognize patterns, make decisions, and even improve their performance over time without explicit human intervention.

➢ **Benefits:**

- Increases efficiency and productivity.
- Reduces human error.
- Automates repetitive tasks.
- Provides insights from large datasets.
- Enhances decision-making processes.

➢ **Challenges:**

- Ethical concerns.
- High development and implementation costs.
- Lack of transparency in some AI models.
- Potential misuse or unintended consequences.

3. PROPOSED SYSTEM

The proposed system is designed to detect driver drowsiness in real-time using video or webcam technology. It focuses on analyzing facial features, such as eye closure, blinking, and yawning, to identify signs of fatigue. The system calculates the Eye Aspect Ratio (EAR) from eye landmarks to determine if the eyes are open or closed and measures the distance between the upper and lower lips to detect yawning. Using Python, OpenCV, and Dlib, the system processes video frames to monitor these indicators. If signs of drowsiness are detected, it triggers an alarm to alert the driver. This non-intrusive approach is cost-effective and provides immediate feedback, addressing challenges like lighting variations and head movement to improve road safety.

How it works?

The device works through a face reading mechanism and has a camera attached to it, where a person's eye movement and the number of yawns they take are monitored before sending alerts.

The cut-off time for a person to blink his eye is 1.5 seconds. If a driver has his eyes shut for more than that, alerts would be sent. The same happens even if he yawns more than 3 times.

4. FEATURES

Here are the key features of the Driver Drowsiness Detection System:

1. Real-Time Monitoring:

- Continuously tracks the driver’s face and behavior using a camera.
- Works in real-time to ensure immediate response.

2. Facial Landmark Detection:

- Uses AI-based image processing to detect eyes, mouth, and head position.
- Identifies signs of drowsiness such as closed eyes, yawning, and head tilting.

3. Eye Aspect Ratio (EAR) & Mouth Aspect Ratio (MAR) Calculation:

- EAR detects eye closure duration to determine drowsiness.
- MAR detects yawning frequency, another key indicator of fatigue.

4. Non-Intrusive Detection:

- No need for physical sensors on the driver.
- Uses a simple camera setup, making it comfortable to use.

5. AI & Machine Learning Integration:

- Uses OpenCV and Dlib for facial recognition.
- Can be enhanced with deep learning (CNNs) for improved accuracy.

6. Alert Mechanism:

- Triggers an alarm (buzzer/speaker) if drowsiness is detected.
- Option for visual alerts (on-screen warnings) or vibrations.

7. Works in Various Lighting Conditions:

- Uses Infrared (IR) cameras to detect drowsiness at night.
- Can adapt to different environmental lighting conditions.

8. Cost-Effective & Easy to Implement:

- Uses low-cost hardware like a USB camera or Raspberry Pi.
- Open-source software tools make it easy to develop and deploy.

9. Compatibility with Vehicle Systems:

- Can be integrated with vehicle control systems for enhanced safety.
- Potential for automatic braking or steering correction in future upgrades.

5. ADVANTAGES & CHALLENGES

5.1 ADVANTAGES:

- **Prevents Accidents:**
  Reduces road accidents caused by driver fatigue.

- **Real-Time Monitoring:**
  Detects drowsiness instantly and alerts the driver.

- **Non-Intrusive Approach:**
  No physical sensors; uses a camera for detection.

- **Cost-Effective:**
  Relies on affordable hardware like a webcam and open-source AI tools.

- **AI-Based Accuracy:**
  Machine learning improves detection accuracy over time.

- **Works in Low Light :**
  Infrared (IR) cameras can detect drowsiness at night.

- **Customizable Alerts:**
  Offers flexible alert options such as alarms, screen warnings, or vibration alerts.

5.2 CHALLENGES:

- **Lighting Conditions:**
  Low-quality cameras may struggle to perform accurately in extreme lighting, such as bright sunlight or low-light environments.

- **Head Movements:**
  Excessive or sudden head movements by the driver can cause face detection algorithms to fail, reducing system reliability.

- **False Positives:**
  The system might incorrectly trigger alerts for normal behaviors like blinking or brief glances away from the road.

- **Hardware Limitations:**
  High accuracy depends on good-quality cameras and sensors, which may not always be available or affordable.

- **Processing Power:**
  AI-based detection systems can be computationally intensive, requiring powerful hardware to process data in real time.

- **User Acceptance :**
  Drivers may find frequent or unnecessary alerts annoying, intrusive, or distracting, leading to resistance in adopting the technology.

6. APPLICATION

Applications of Driver Drowsiness Detection System are:

- **Road Safety Enhancement**
  - Reduces accidents caused by driver fatigue.
  - Improves overall traffic safety.

- **Commercial Vehicle Monitoring**
  - Used in trucks, taxis, and buses to enhance driver safety.
  - Helps fleet management companies monitor driver alertness.

- **Automobile Industry Integration**
  - Can be integrated into modern Advanced Driver Assistance Systems (ADAS).
  - Potential for use in self-driving technologies.

- **Public Transport Safety**
  - Ensures bus and train drivers remain alert during operations.
  - Prevents accidents in long-distance travel scenarios.

- **Insurance & Risk Management**
  - Insurance companies can assess driver safety using this system.
  - Helps reduce accident-related claims.

- **Military & Defense**
  - Ensures soldiers and pilots stay alert during critical missions.
  - Can be integrated into military transport systems for enhanced safety.

- **Smart Cities & Traffic Systems**
  - Can be incorporated into smart transportation networks.
  - Contributes to intelligent traffic management solutions.

7. COMPONENTS

The working of the Driver Drowsiness Detection System involves several steps, combining image processing and real-time analysis to monitor the driver's state of alertness. Here's how it works:

7.1 HARDWARE COMPONENTS

- **Infrared (IR) Cameras:**

A webcam or camera is installed in front of the driver to continuously capture video frames of the driver's face. A camera with infrared LEDs are commonly used to improve the night vision.

Figure 1: USB Infrared (IR) Night Vision Camera

- **Processing Unit:**

Devices like **Raspberry Pi** or **NVIDIA Jetson Nano** for edge computing.

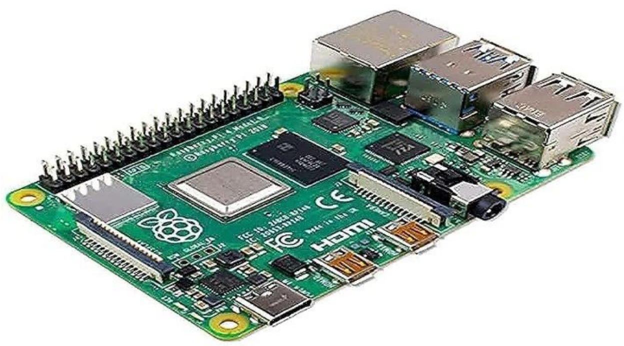

Figure 2: Raspberry Pi -4

➢ **Alert System:**

Emits a sound when drowsiness is detected to alert the driver.

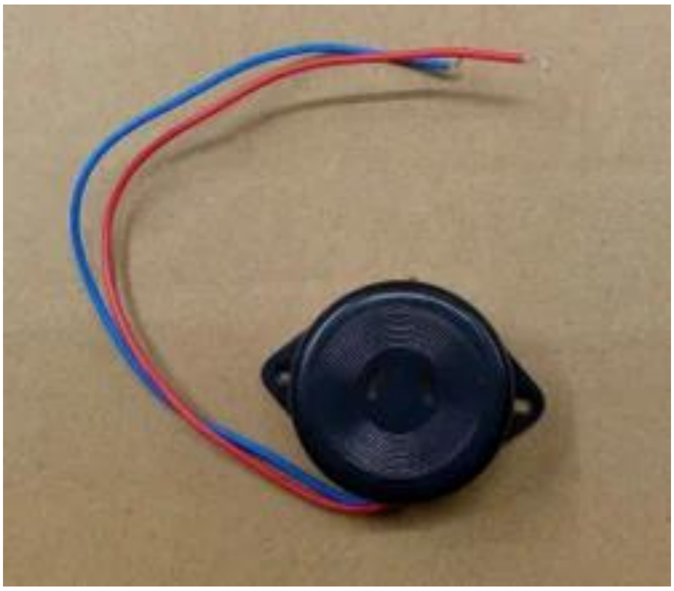

Figure 3: Buzzer

➢ **Power Supply:**

- **Car USB Port:**

Supplies 5V power directly to the camera and processing unit.

Figure 4: Car USB port

- **External Power Bank:**

For portability and independent operation.

7.2 SOFTWARE COMPONENTS

- **Programming Framework:**
  - **Python:**

    Primary language for implementing the detection algorithms.
  - **Libraries include:**
    - **OpenCV**: For image processing and face detection.
    - **Dlib** : For facial landmark detection and tracking.
    - **Numpy** : For mathematical calculations like EAR and MAR.
    - **TensorFlow/Keras** : For deep learning-based classification.

- **Face Detection:**

  HOG + SVM or Haar Cascade Classifier: Detects the driver's face from the video frames.

- **Facial Landmark Detection:**

  Dlib's 68-point Facial Landmark Detector: Identifies critical points on the face such as:
  - Eyes (landmarks 36–47)
  - Mouth (landmarks 48–67)

- **Eye Aspect Ratio (EAR):**
  - Calculate the ratio of vertical and horizontal distances between eye landmarks:
  - Detects eye closure when EAR falls below a threshold (EAR<0.25).

- **Mouth Aspect Ratio (MAR):**
  - Measures the vertical distance between the upper and lower lips relative to the horizontal distance
  - Detects yawning based on increased MAR (MAR > 0.6).
- **Threshold-Based Drowsiness Detection:**
  - Combines EAR and MAR results to identify drowsiness.
  - Triggers alerts if thresholds (e.g., EAR < 0.25 or MAR > 0.6) are consistently exceeded.

8. SYSTEM ARCHITECTURE

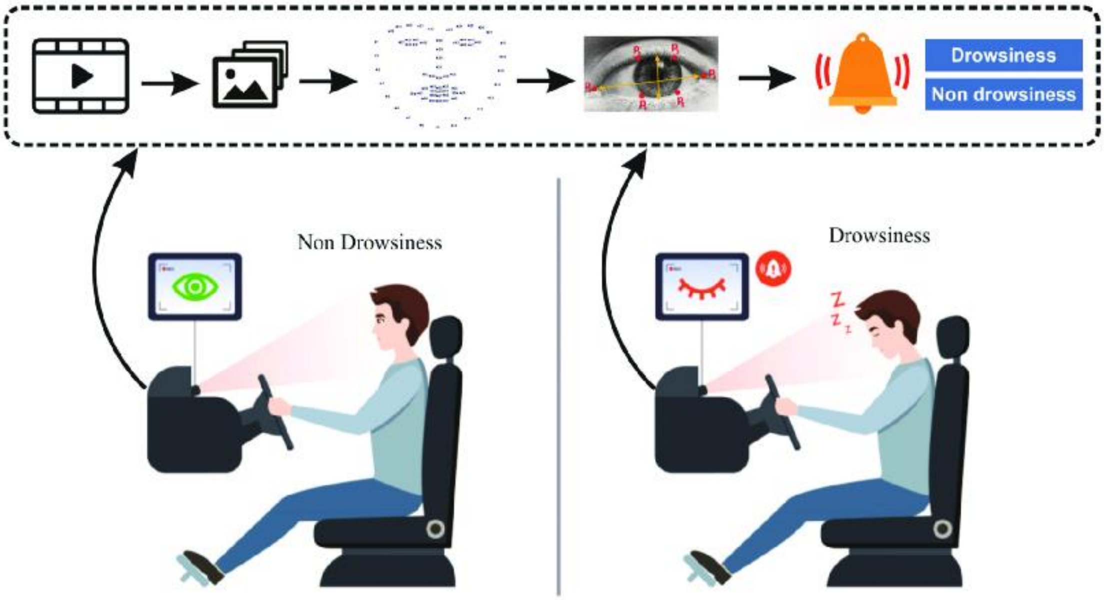

Figure 5: Driver drowsiness detection system based on Facial recognition & Eyes Detection

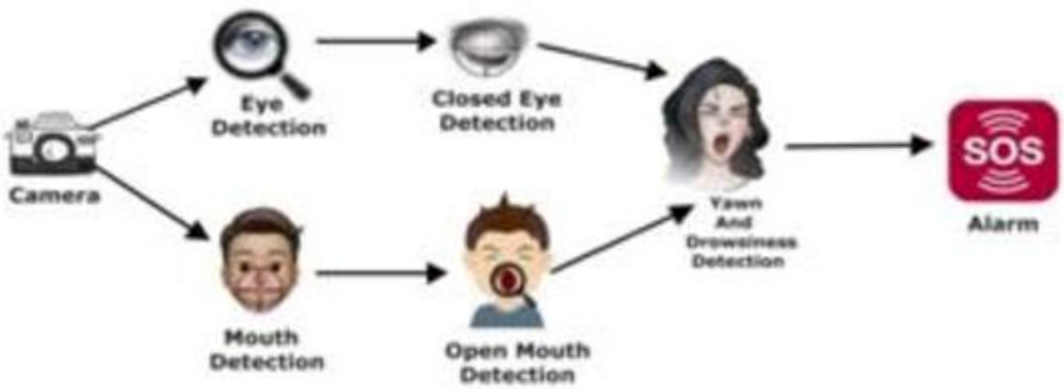

Figure 6: Driver drowsiness detection system base on Yawn detection

9. SYSTEM FLOWCHART

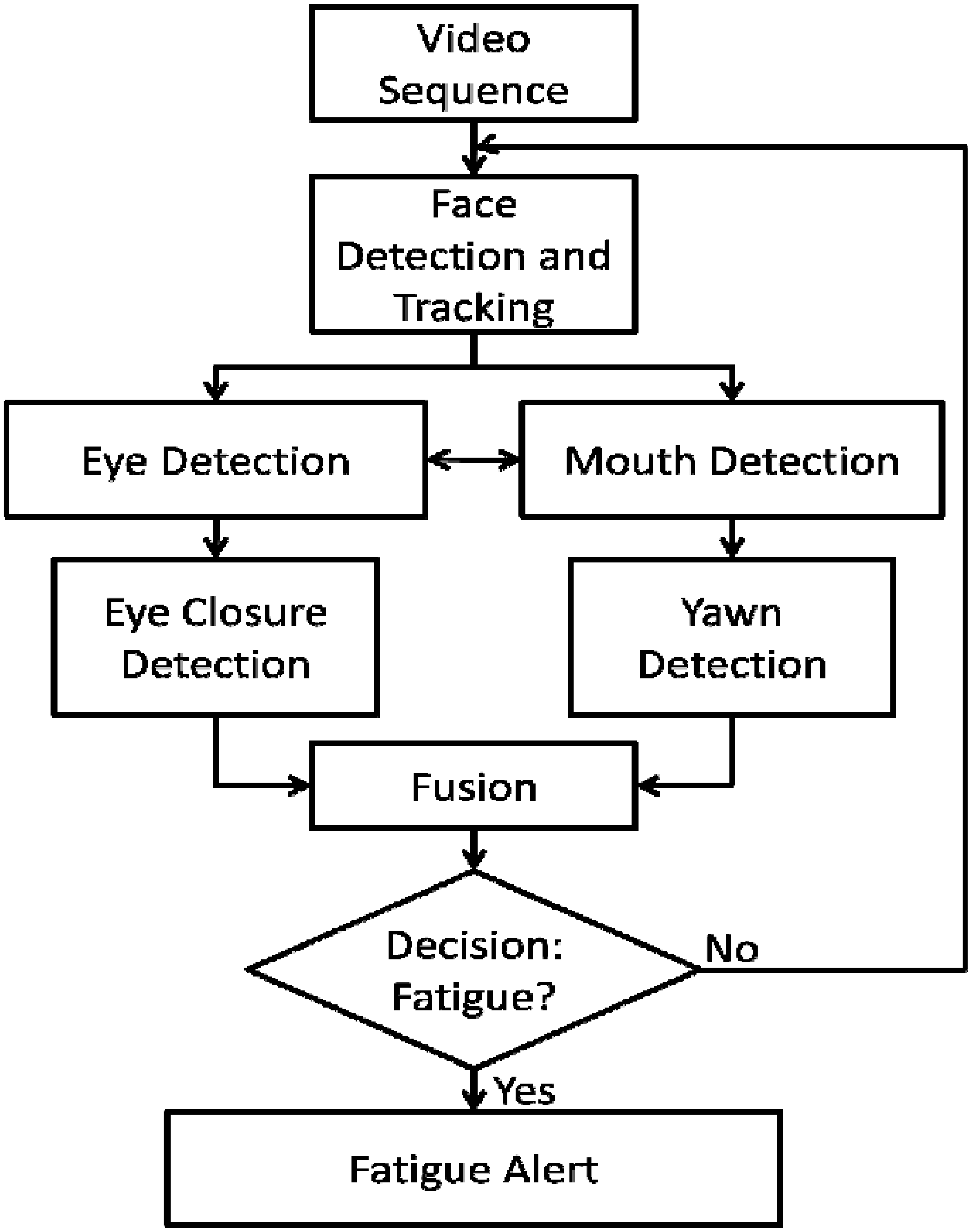

Figure 7: Driver drowsiness detection system flowchart

10. BLOCK DIAGRAM

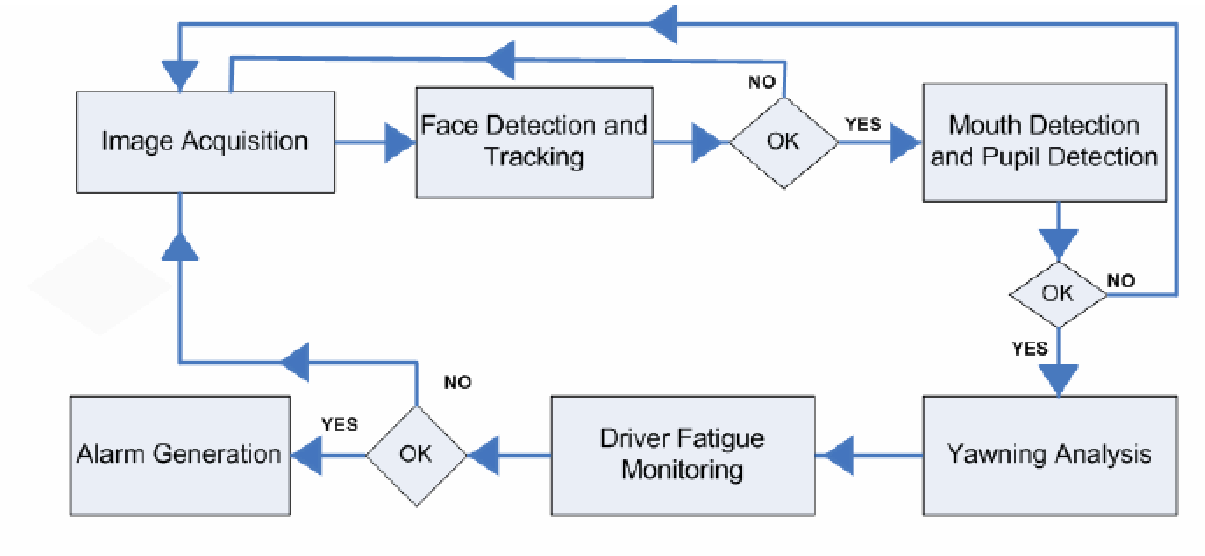

Figure 8: Driver drowsiness detection system -Block diagram

10.1 USE CASE DIAGRAM

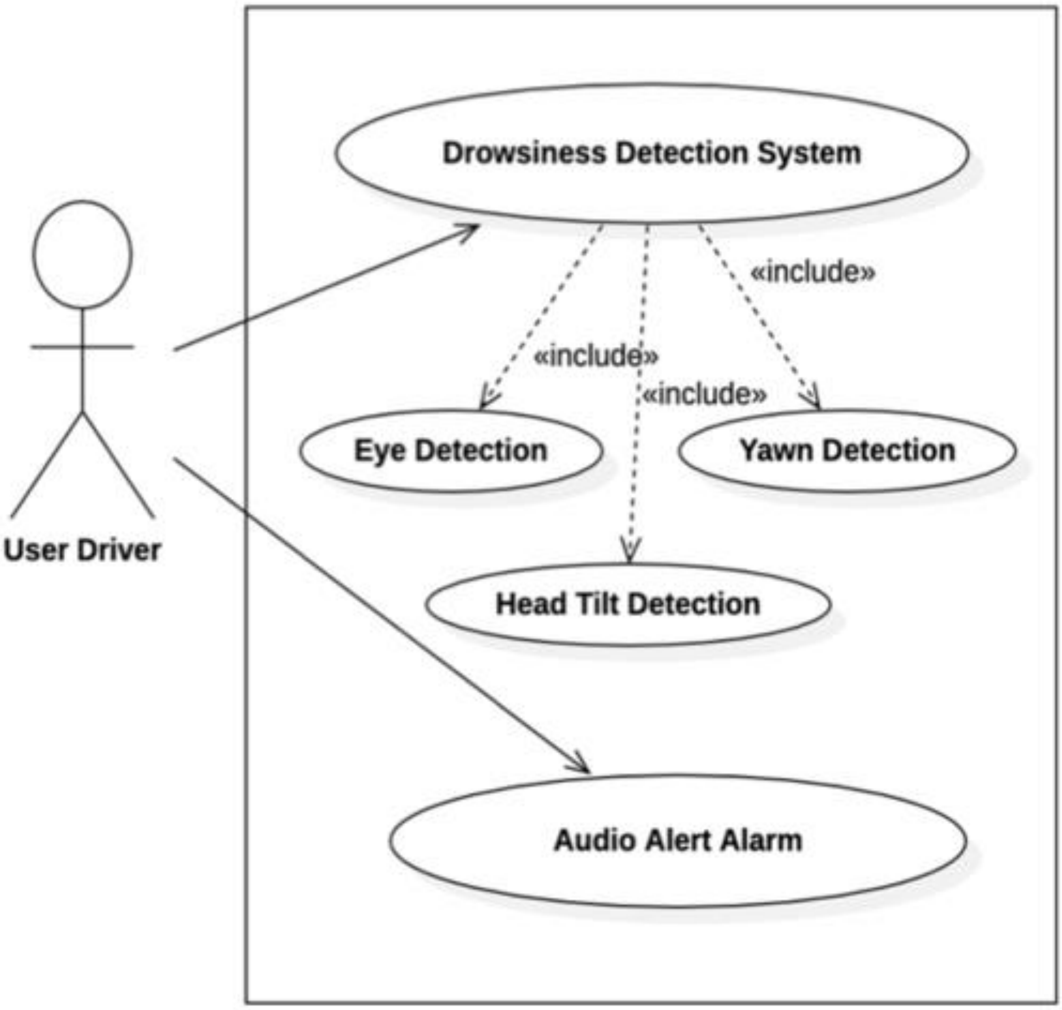

Figure 9: Driver drowsiness detection system use case-diagram

11. OBJECTIVES

Here we focus on t various objectives, which are:

- To suggest ways to detect drowsiness while driving.
- To observe eyes and mouth from the video images of participants in the experiment of driving simulation conducted by MIROS that can be used as an indicator of drowsiness.
- To investigate the physical changes that take place during drowsiness.
- To develop a system that use eyes closure and yawning as a way to detect drowsiness.

12.CONCLUSION

The Driver Drowsiness Detection System is an essential technological advancement aimed at reducing road accidents caused by driver fatigue. By leveraging image processing, artificial intelligence, and facial landmark detection, the system effectively monitors a driver’s alertness in real-time. It identifies eye closure, and yawning using Eye Aspect Ratio (EAR) and Mouth Aspect Ratio (MAR) calculations to determine drowsiness.

This project successfully demonstrates a non-intrusive, cost-effective, and efficient solution for detecting driver fatigue. The use of OpenCV, Dlib, and Python-based algorithms ensures high accuracy in tracking facial features. Moreover, infrared cameras enhance functionality even in low-light conditions, making the system suitable for various driving environments.

However, challenges such as lighting variations, head movements, and false detections need to be addressed. Future enhancements can include deep learning-based models, integration with vehicle automation, and improved sensor technology to increase accuracy and reliability.

In conclusion, this system offers a practical and effective approach to improving road safety by reducing fatigue-related accidents. With continuous advancements in AI and computer vision, driver monitoring systems will play a significant role in the future of intelligent transportation and vehicle safety systems.

13.REFERENCES

1. Ralph Oyini Mbouna, Seong G. Kong, Senior Member, IEEE, (2013), Visual Analysis of Eye State and Head Pose for Driver Alertness Monitoring, (IEEE), pp.1462-1469, vol.14, USA
2. Xiao F., Bao C.Y., Yan F.S. published "Yawning detection based on gabor wavelets and LDA." J. Beijing Univ. Technol. 2009; 35:409–413. [Google Scholar]
3. Rau P. “Driver Drowsiness Detection and Warning System for Commercial Vehicle” “Driver’s Field Operational Test Design, Analysis, and Progress”.
4. “National Highway’s Traffic Safety Administration; Washington, DC, USA: 2005” [Google Scholar]
5. Boon-Giin Lee and Wan-Young Chung published “Driver Alertness Monitoring Using Fusion of Facial Features and Bio-Signals”, (IEEE) Sensors journal, vol. 12, no. 7,2012
6. “Detecting Driver Drowsiness Based on Sensors A Review”,pp.16937-16953, ISSN 1424-8220, Malaysia 2012.
7. Research Gate : figure 5- 9 used in this report where from this research gate website.